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Master IASD — Intelligence Artificielle et Systèmes Décisionnels

PhishGuard

A Multilingual RAG-Based Phishing Detection System

Détection et Analyse du Phishing · English / Français / Arabic

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1. Introduction

Phishing attacks represent one of the most pervasive threats in modern cybersecurity, exploiting human psychology through deceptive messages across email, SMS, social media, and malicious URLs. Traditional rule-based detection systems fail to adapt to evolving attack patterns and lack the ability to explain their decisions to end users.

This project addresses these limitations through **PhishGuard**, a complete RAG-based system that:

- Detects phishing in **three languages** (EN, FR, AR)
- Analyzes **four content channels** (email, SMS, URL, social media)
- Provides **explainable decisions** with suspicious element identification
- Extends knowledge via **PDF upload** capability

The project strictly adheres to the course requirements: all data was manually constructed, no public datasets were used, and the LLM is exclusively integrated within the RAG pipeline architecture.

2. Dataset Construction

2.1. Data Collection Strategy

The dataset was built entirely from scratch as required, combining manual creation with automated collection from verified sources. Four complementary channels were covered to ensure diversity and realistic representation of real-world phishing scenarios.

Table 1: Dataset composition by channel

Channel	Total	Phishing	Legit
URL	1,264	438	826
SMS	856	426	430
Email	644	290	354
Social Media	274	205	69
Total	**3,038**	**1,359**	**1,679**

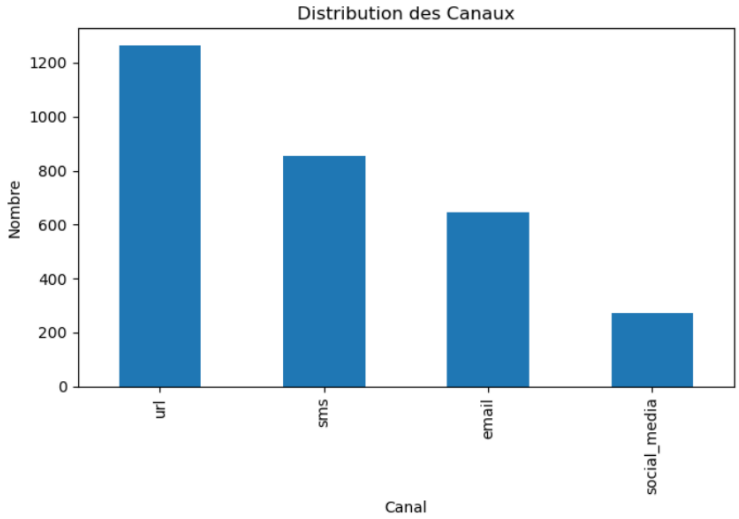

Figure 1: Distribution des messages par canal

URL data: Phishing URLs were collected from *PhishTank* (verified, currently active submissions), filtered to exclude entries with unknown targets, capped at 20 samples per brand to ensure diversity. Legitimate URLs were sourced from the *Majestic Million* top-ranked domains, filtered by TLD (.com, .fr, .ma) for language coverage.

SMS & Email data: Messages were manually authored by team members, covering realistic scenarios including bank notifications, delivery alerts, prize scams, and credential theft attempts across three languages.

Social media data: Suspicious and legitimate posts were manually constructed to reflect realistic social engineering patterns observed on platforms like Facebook and WhatsApp.

2.2. Data Schema

Each entry follows a standardized 8-column schema:

Table 2: Dataset schema

Field	Values
`text`	Raw message content
`sender`	Email / phone / null
`label`	phishing / legitimate
`channel`	email, sms, url, social_media
`language`	en, fr, ar
`attack_type`	8 categories (see below)
`risk_level`	low / medium / critical

Attack types defined: `credential_theft`, `brand_impersonation`, `urgency`, `prize_scam`, `threat`, `typosquatting`, `malware_link`, `none`.

2.3. Language Distribution

Table 3: Dataset language distribution

Language	Count	%
English (en)	1,508	49.6%
Arabic (ar)	899	29.6%
French (fr)	631	20.8%
Total	**3,038**	**100%**

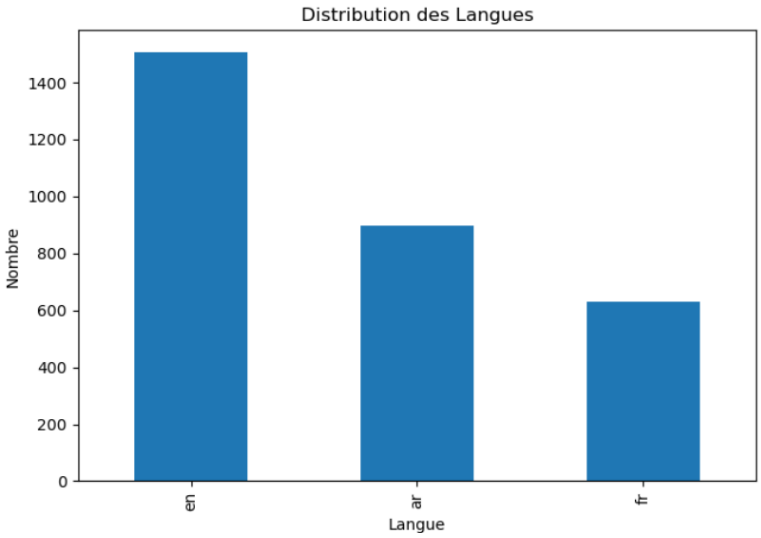

Figure 2: Distribution des langues dans le dataset

2.4. Synthetic Data Generation

As required by the course guidelines, automatic synthetic data generation was implemented to enrich the dataset. Two complementary approaches were developed:

2.4.1. Template-Based Generation

A `synthetic_data_generator.py` script was created using parameterized message templates filled with domain-specific word banks covering brands (PayPal, CIH Bank, Orange Maroc, etc.), action verbs (suspended, blocked, expired), urgency markers, and realistic link patterns. Templates were designed for all three languages and all non-URL channels, producing **432 new synthetic rows**.

2.4.2. NLP Augmentation

A separate `augmentation.py` script applies five NLP augmentation techniques to 30 selected messages (15 phishing + 15 legitimate), generating 5 variations each for a total of **150 augmented rows**:

1. **Synonym replacement** — WordNet-based word substitution
2. **Random word swap** — position interchange of two tokens

3. **Random deletion** — probabilistic word removal ($p = 0.15$)
4. **Random insertion** — synonym injection at random positions
5. **Character noise** — realistic typo simulation ($p = 0.05$)

2.5. Preprocessing Pipeline

A light preprocessing pipeline (`preprocessing.py`) was applied to produce the `preprocessed_text` column used by the retrieval module, while preserving the original `text` column for LLM context and dashboard display. Steps applied:

1. HTML tag removal
2. Repeated punctuation normalization
3. Whitespace normalization
4. Selective lowercasing (URL tokens preserved)
5. Stopword removal (NLTK: EN, FR; custom list: AR)

URLs, email addresses, numbers, and verification codes were explicitly preserved as they constitute critical phishing indicators.

3. LLM

3.1. LLM Models Tested

3.1.1. LLaMA 3.3-70B (llama-3.3-70b-versatile)

LLaMA 3.3-70B is Meta AI's large-scale model, accessible via the Groq API. Its 70 billion parameters provide superior contextual understanding and reasoning capabilities.

3.1.2. LLaMA 3.1-8B (llama-3.1-8b-instant)

LLaMA 3.1-8B is the lightweight version of LLaMA 3, optimized for low latency ("instant"). With 8 billion parameters, it offers an excellent performance-to-speed ratio.

3.1.3. Qwen3-32B (qwen/qwen3-32b)

Qwen3-32B is a 32-billion-parameter model developed by Alibaba Cloud, featuring an internal "thinking" mechanism.

3.2. Prediction Distribution

Table 4: Prediction distribution by model

Model	Predicted Legitimate	Predicted Phishing	Total
LLAMA70B	37	63	100
LLAMA8B	25	75	100
Qwen32B	6	94	100

3.3. Global Results

Table 5: Global results of LLM models

Model	Accuracy	Precision	Recall	F1 Phishing	F1 Legit	F1 Macro	Avg. Time (s)
LLAMA 3.3-70B	**77.00%**	**71.43%**	**90.00%**	**79.65%**	**73.56%**	**76.60%**	**2.15**
LLAMA 3.1-8B	71.00%	64.00%	96.00%	76.80%	61.33%	69.07%	0.82
Qwen3-32B	56.00%	53.19%	100.00%	69.44%	21.43%	45.44%	1.94

3.4. Confusion Matrices

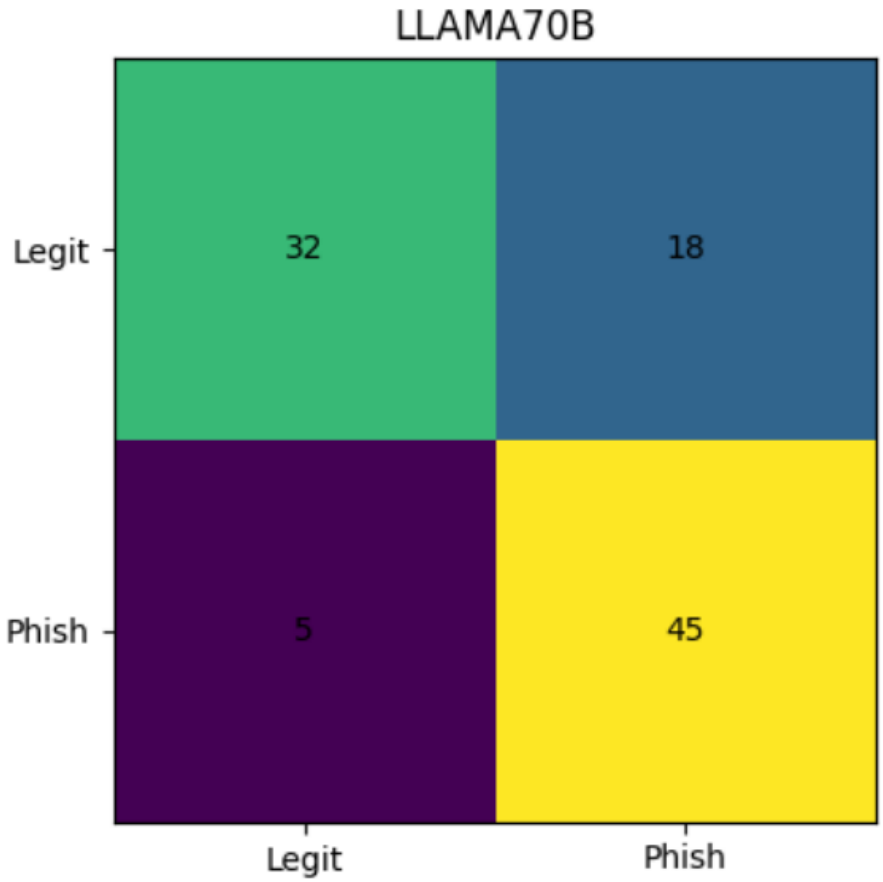

Figure 3: Confusion matrix — LLAMA 3.3-70B (Acc=77%, Recall=90%)

Interpretation: The confusion matrix of LLAMA 3.3-70B reveals a good balance between phishing detection and the preservation of legitimate messages. Out of 100 tested messages (50 legitimate, 50 phishing), the model correctly identified 32 legitimate messages (true negatives) and 45 phishing messages (true positives). Errors break down into 18 false positives (legitimate messages classified as phishing) and 5 false negatives (phishing messages not detected). The false negative rate of only 10% (5/50) is excellent for a security system, as it minimizes the risk of missing an attack. The 18 false positives (36% of legitimate messages) represent an acceptable inconvenience for the user, who can dismiss an alert on a legitimate message.

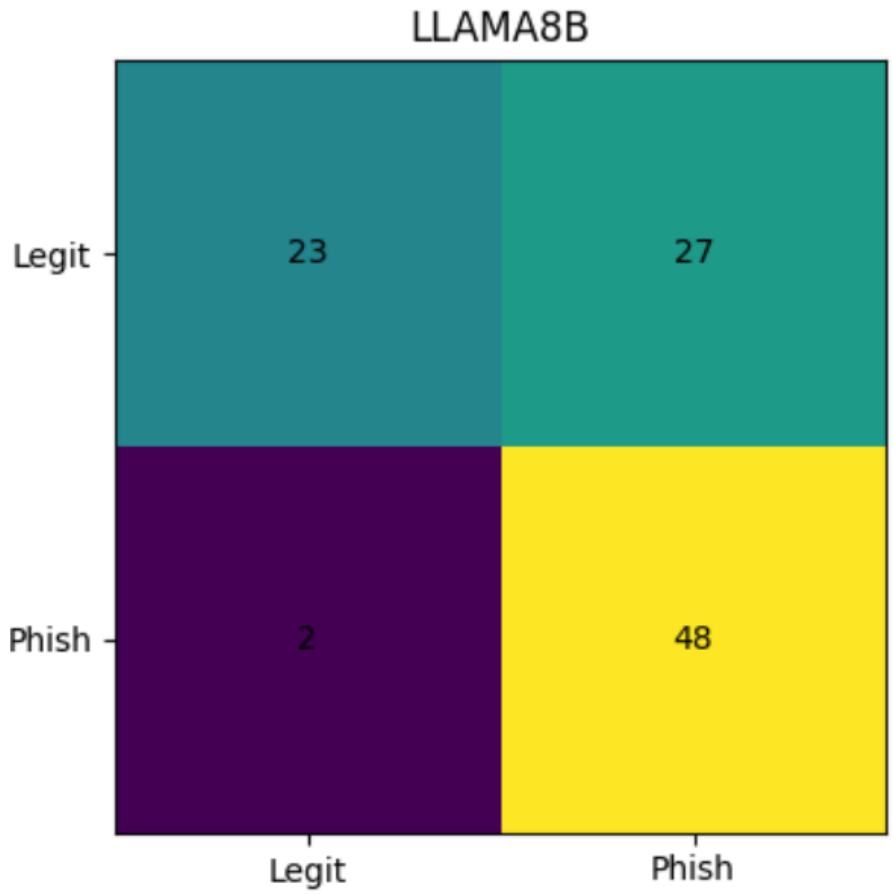

Figure 4: Confusion matrix — LLAMA 3.1-8B (Acc=71%, Recall=96%)

Interpretation: LLAMA 3.1-8B shows a strong bias toward phishing detection, with only 2 false negatives (4% error rate) but 27 false positives (54% of legitimate messages). This model is therefore highly sensitive (it detects almost all phishing) but poorly specific (it flags too many legitimate messages as suspicious). This behavior is explained by its reduced size (8B parameters), which makes it less capable of nuanced predictions. Although its speed (0.82 s) is appealing, its high false positive rate makes it unsuitable for standalone use, as users would eventually start ignoring alerts ("alarm fatigue").

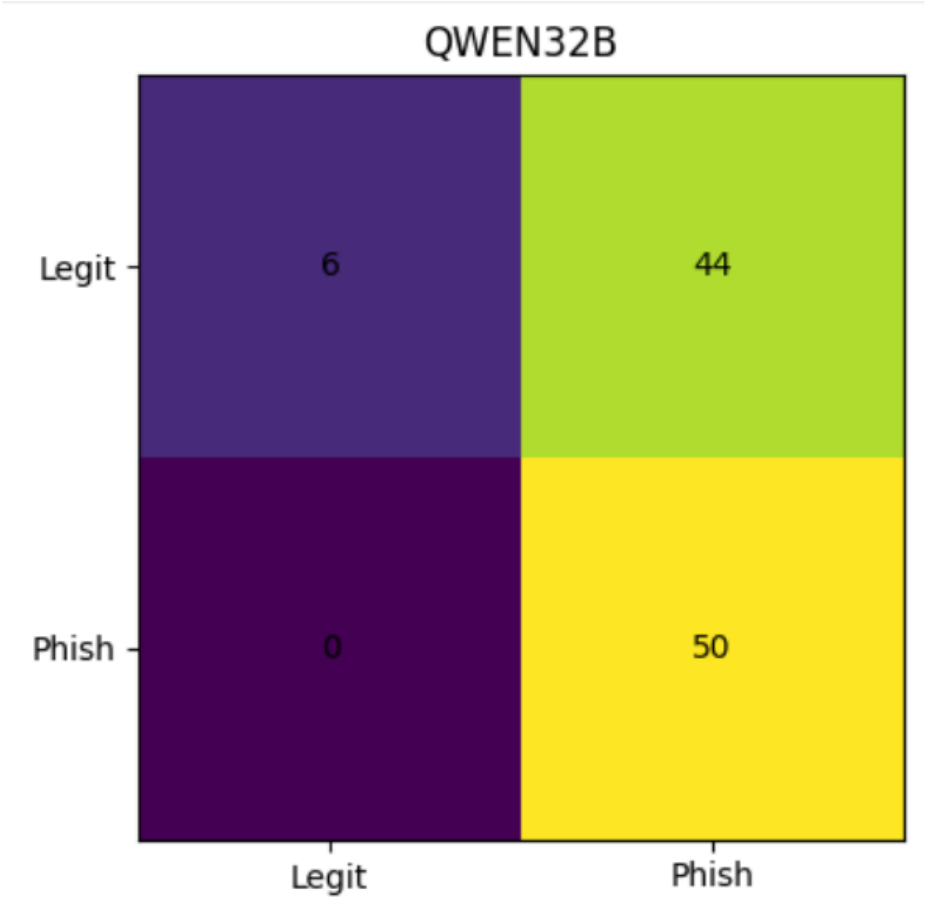

Figure 5: Confusion matrix — Qwen3-32B (Acc=56%, Recall=100%)

Interpretation: Qwen3-32B exhibits extreme behavior: 0 false negatives (all phishing detected) but 44 false positives (88% of legitimate messages classified as phishing). This systematic bias toward the phishing class is likely due to its internal "thinking" mechanism (`<think>...</think>`), which encourages hypervigilance. While the perfect recall (100%) may seem attractive, the operational cost of 44 false positives out of 50 legitimate messages is prohibitive: users would receive an alert on almost every message, rendering the system unusable.

3.5. Comparative Performance Chart

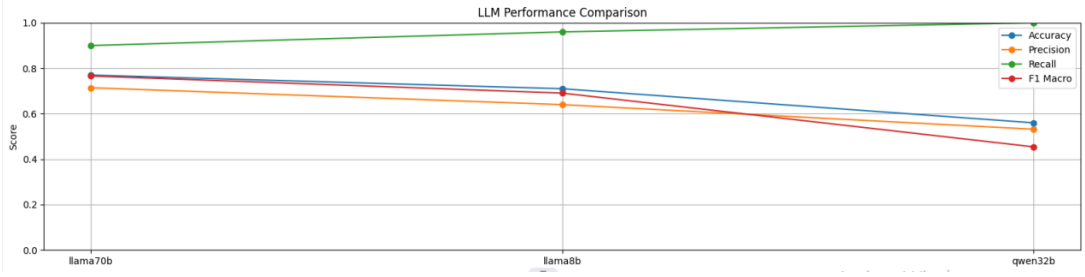

Figure 6: LLM performance comparison and confusion matrices

Interpretation: This comparative chart summarizes the performance of the three models across four key metrics. LLAMA 3.3-70B leads on accuracy (77%), precision (71.43%), and F1 Macro (76.60%), confirming its superior balance. LLAMA 3.1-8B excels on recall (96%) but at the cost of lower precision. Qwen3-32B achieves perfect recall (100%) but the lowest precision (53.19%). The F1 Macro, as the harmonic mean of precision and recall, is the most representative metric for this binary problem. With 76.60%, LLAMA 70B clearly outperforms LLAMA 8B (69.07%) and Qwen32B (45.44%), visually justifying the choice of LLAMA 3.3-70B as the primary model.

3.6. Performance by Language

Table 6: Performance by language

Model	Language	Samples	Accuracy	F1 Macro
LLAMA70B	EN	55	81.82%	79.35%
LLAMA70B	AR	30	76.67%	76.00%
LLAMA70B	FR	15	60.00%	59.82%
LLAMA8B	EN	55	83.64%	81.14%
LLAMA8B	AR	30	56.67%	55.43%
LLAMA8B	FR	15	53.33%	53.33%
Qwen32B	EN	55	70.91%	57.36%
Qwen32B	AR	30	40.00%	32.50%
Qwen32B	FR	15	33.33%	30.56%

Interpretation: The per-language analysis reveals significant disparities. For LLAMA 70B, performance is excellent in English (accuracy 81.82%, F1 Macro 79.35%), good in Arabic (76.67%, 76.00%), but poor in French (60.00%, 59.82%). This performance drop in French is explained by the underrepresentation of this language in the training dataset (631 messages vs. 1,508 for English), and potentially in the LLM's own pretraining data. For LLAMA 8B, the gap is even more pronounced: English reaches 83.64% accuracy while Arabic and French drop to 56.67% and 53.33% respectively. Qwen32B struggles particularly on Arabic (40%) and French (33.33%). These results highlight the need to enrich the French dataset and to consider a more balanced multilingual fine-tuning approach.

4. System Architecture

4.1. RAG Pipeline Overview

The complete pipeline follows six sequential steps:

RAG Pipeline Flow

1. **Input** — User submits message or URL
2. **Preprocessing** — Light text normalization
3. **Encoding** — Sentence-BERT vectorization
4. **Retrieval** — Hybrid FAISS + BM25 with RRF
5. **Prompt Construction** — Context-enriched prompt
6. **Generation** — LLM classification + explanation

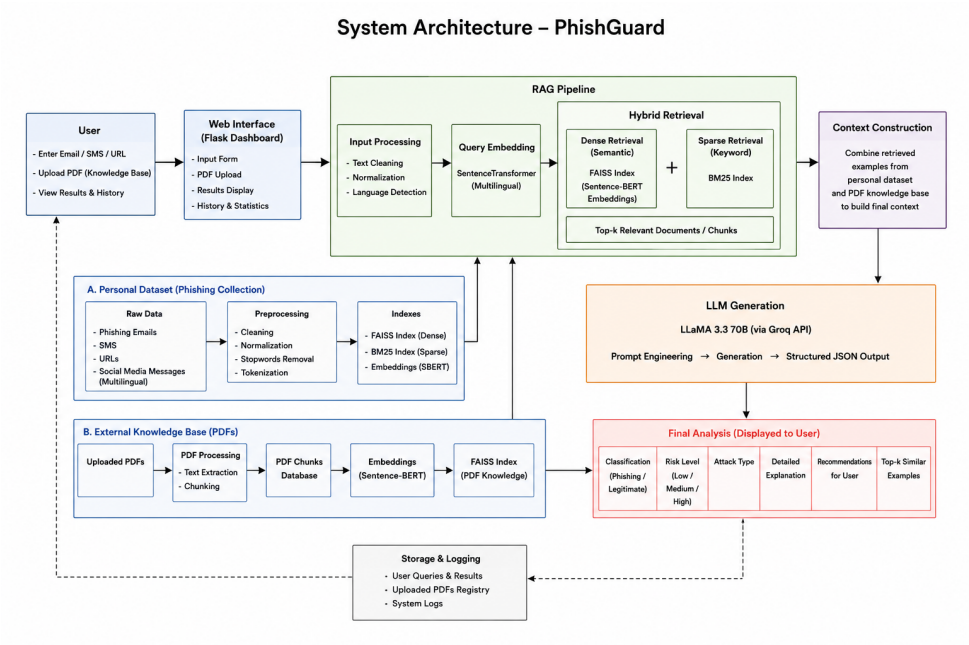

Figure 7: System architecture diagram

4.2. URL Analysis Module

A dedicated URL analyzer (`url_analyzer.py`) extracts 13 features from submitted URLs before passing them to the RAG pipeline as structured text:

- URL length (> 75 characters flagged)
- Subdomain count (> 2 flagged)
- Presence of IP address in domain
- Suspicious TLD detection (.xyz, .tk, .ml, .cf)
- Suspicious keyword matching (verify, secure, login, etc.)
- HTTP vs HTTPS protocol check
- Dash count in domain
- `@` symbol presence

4.3. PDF Knowledge Base

The system supports dynamic knowledge extension through PDF upload (`pdf_knowledge.py`). Uploaded documents are processed as follows:

1. Text extraction via `PyMuPDF (fitz)`
2. Chunking into 500-character segments
3. Embedding with *paraphrase-multilingual-MiniLM-L12-v2*

4. Indexing into a dedicated FAISS index
5. Registry persistence via JSON

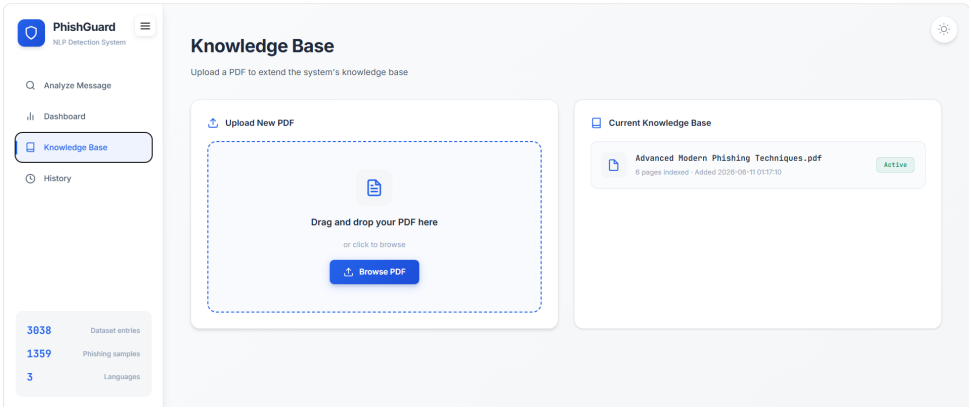

Figure 8: Knowledge Base management interface

5. Retrieval Module

5.1. Architecture

The retrieval module (`retrieval_module.py`) implements a hybrid search strategy combining dense semantic retrieval with sparse keyword matching:

FAISS (Dense Retrieval): The `IndexFlatIP` index with L2 normalization enables cosine similarity search over sentence embeddings, providing semantic understanding of message intent beyond keyword overlap.

BM25 (Sparse Retrieval): The `BM25Okapi` implementation captures exact keyword matches critical for phishing detection (brand names, suspicious verbs, urgency markers).

Reciprocal Rank Fusion (RRF): Both result sets are fused using the RRF algorithm with $k = 60$, selecting from the top-20 candidates of each method before returning the final Top-k results. This approach consistently outperforms single-method retrieval.

5.2. Embedding Model Benchmarking

Three multilingual embedding models were evaluated on 100 queries sampled from the dataset across all three languages:

Table 7: Retrieval model benchmarking results

Model	HR@5	MRR	nDCG@5	P@5	Time (ms)
all-MiniLM-L6-v2	**0.98**	**0.899**	0.807	0.798	**13.0**
distiluse-multilingual-v2	0.97	0.870	**0.809**	0.808	35.3
multilingual-MiniLM-L12-v2	0.94	0.858	0.792	0.784	24.7

Selected model: *all-MiniLM-L6-v2* was selected for production due to its best Hit Rate@5 (0.98), highest MRR (0.899), and fastest search time (13ms average), making it optimal for real-time web application deployment.

6. LLM Generation Module

6.1. Prompt Engineering

The generation prompt was carefully engineered to guide the LLM through a structured analysis process:

Listing 1: RAG prompt structure

```
You are a cybersecurity expert specialized
in phishing detection.

Use the retrieved examples as context.

Retrieved Examples:
{context}  # Top-5 similar cases

Analyze the following message.
Before classifying, check for:
1. Urgency language
2. Suspicious links or domains
3. Brand impersonation
4. Credential requests
5. Prize/reward scams

Return ONLY valid JSON:
{
  "label": "phishing|legitimate",
  "confidence": <float 0.0-1.0>,
  "risk_score": <int 0-100>,
  "language": "en|fr|ar",
  "attack_type": "credential_theft|...",
  "explanation": "...",
  "suspicious_elements": [...],
  "recommendation": "..."
}
```

Key design decisions:

- **Chain-of-thought prompting** — explicit checklist forces structured reasoning before classification
- **Structured JSON output** — enables reliable parsing and dashboard integration
- **temperature=0** — deterministic outputs for reproducibility
- **System prompt** — role separation improves instruction following

6.2. LLM Benchmarking

Three models available via the Groq API were benchmarked on 100 balanced samples (50 phishing + 50 legitimate):

Table 8: LLM benchmarking results (100 samples)

Model	Acc.	Prec.	Rec.	F1-Ph.	F1-Leg.	F1-Mac.
LLaMA-3.3-70B	**0.77**	**0.714**	0.90	**0.797**	**0.736**	**0.766**
LLaMA-3.1-8B	0.71	0.640	0.96	0.768	0.613	0.691
Qwen3-32B	0.56	0.532	**1.00**	0.694	0.214	0.454

Analysis: LLaMA-3.3-70B achieves the best balanced performance with 77% accuracy and F1-Macro of 0.766. LLaMA-3.1-8B shows acceptable performance with faster response time (2.55s). Qwen3-32B demonstrates severe bias toward phishing classification (100% recall, near-zero precision for legitimate), likely due to its `<think>` reasoning mode conflicting with strict JSON output requirements.

6.3. Multilingual Performance

Table 9: Per-language accuracy breakdown

Model	EN	AR	FR	Avg
LLaMA-3.3-70B	**81.8%**	**76.7%**	**60.0%**	**72.8%**
LLaMA-3.1-8B	83.6%	56.7%	53.3%	64.5%
Qwen3-32B	70.9%	40.0%	33.3%	48.1%

All models perform worst on French, which is attributed to the relatively smaller French sample size in the benchmark (15 samples) and the dataset (631 FR entries vs 1,508 EN). This is a noted limitation to address in future work.

Selected model: LLaMA-3.3-70B (`llama-3.3-70b-versatile`) was selected for the production RAG pipeline due to superior balanced performance across all three languages.

7. System Modeling

7.1. Use Case Diagram

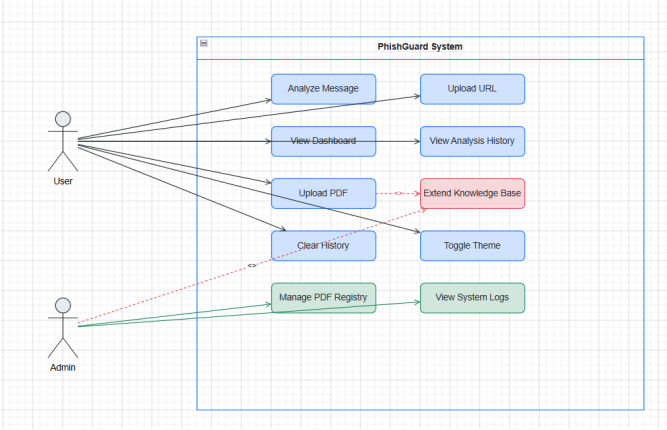

Figure 9: Use case diagram of the PhishGuard system

The use case diagram presents the interactions between the user and the PhishGuard system. The user can submit suspicious messages for phishing analysis, upload cybersecurity PDF documents to enrich the knowledge base, visualize dataset statistics through the dashboard, and consult previous analysis results stored in the history module. The administrator role extends the PDF upload functionality to manage the knowledge base registry and access system logs for monitoring purposes.

7.2. Activity Diagram

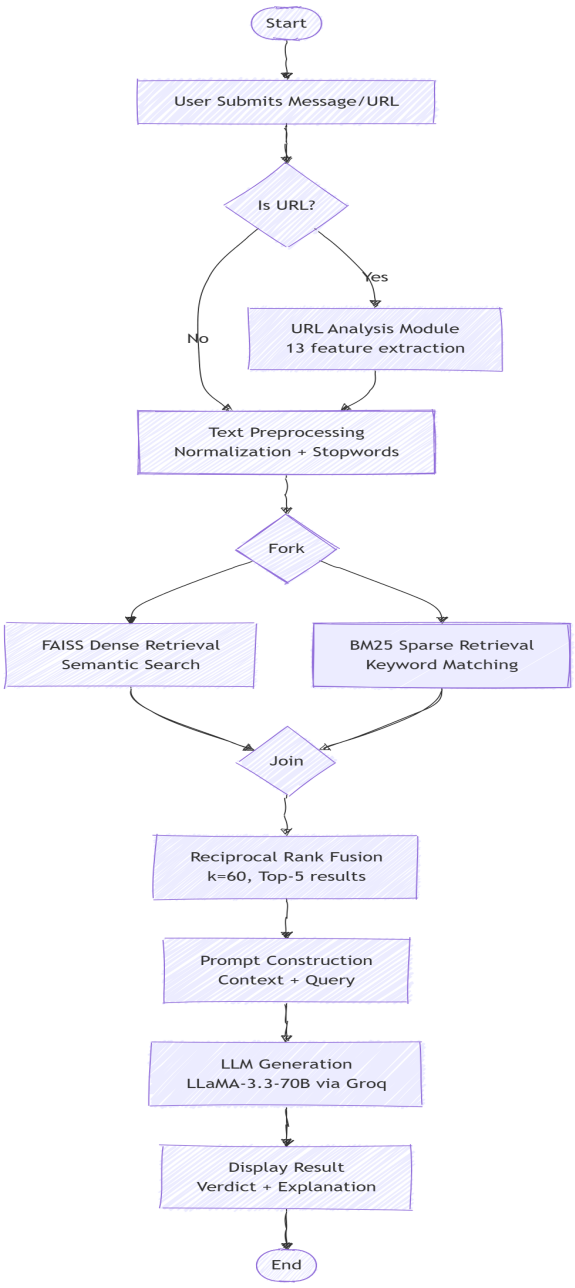

Figure 10: Activity diagram illustrating the phishing analysis workflow using the RAG pipeline

This activity diagram describes the complete processing pipeline followed by PhishGuard. After receiving a suspicious message from the user, the system performs text preprocessing and executes a hybrid retrieval process combining semantic search with FAISS embeddings and keyword-based search using BM25 on the personal phishing dataset. Additionally, specialized information can be retrieved from the PDF knowledge base. The retrieved information is then integrated into an enriched context provided to the LLM, which generates the final analysis including the classification, risk level, explanation of suspicious elements, and security recommendations.

7.3. Component Architecture Diagram

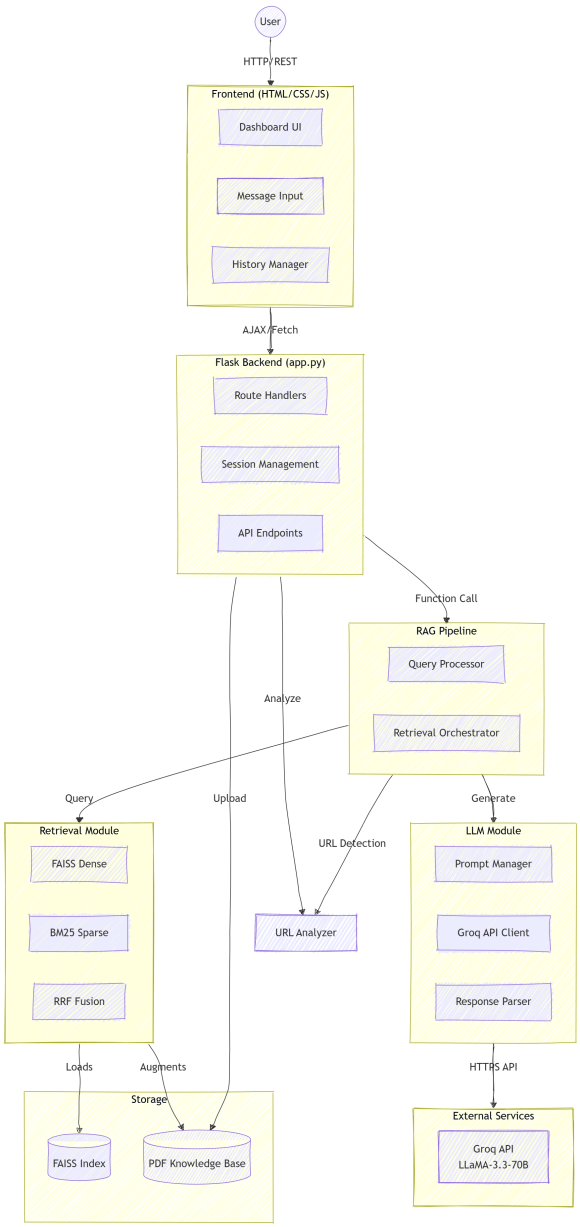

Figure 11: Component architecture of the PhishGuard RAG-based phishing detection system

The component architecture represents the main modules composing the PhishGuard application. The web dashboard, developed with Flask, serves as the interaction layer between the user and the backend services. The system integrates two independent knowledge sources: the personal phishing dataset indexed using FAISS and BM25, and an additional PDF knowledge base indexed separately to preserve the original retrieval logic. The retrieved context is processed by the context builder and transmitted to the LLaMA 3.3 model through the Groq API to produce the final phishing detection result, including the classification, confidence level, explanation, detected attack indicators, and recommendations.

8. Web Dashboard

8.1. Technical Stack

The web application is built with:

- **Backend:** Flask 3.0 (Python)
- **Frontend:** HTML5, CSS3, Vanilla JavaScript
- **Fonts:** Space Grotesk + JetBrains Mono
- **Theme:** Light/Dark toggle with localStorage persistence

8.2. Dashboard Sections

The interface provides four main sections accessible via a collapsible sidebar:

8.2.1. Analyze Message

The main analysis interface accepts any message or URL input. It displays:

- **Verdict** — Phishing / Legitimate with color coding
- **Risk level** — Low / Medium / Critical badge
- **Confidence score** — LLM confidence percentage
- **Language detected** — EN / FR / AR
- **Explanation** — Natural language justification
- **Suspicious elements** — Itemized indicators
- **Recommendations** — Actionable user guidance
- **Similar cases** — Top-5 retrieved dataset examples

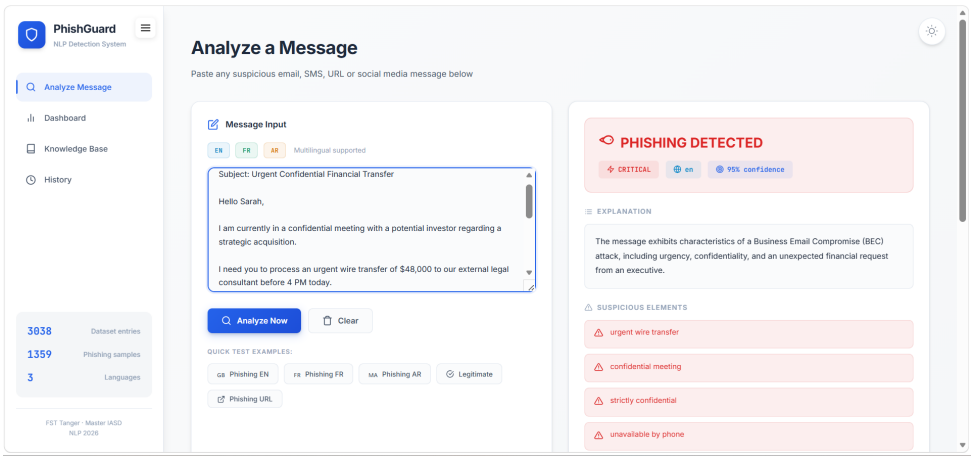

Figure 12: Message analysis result — phishing detection

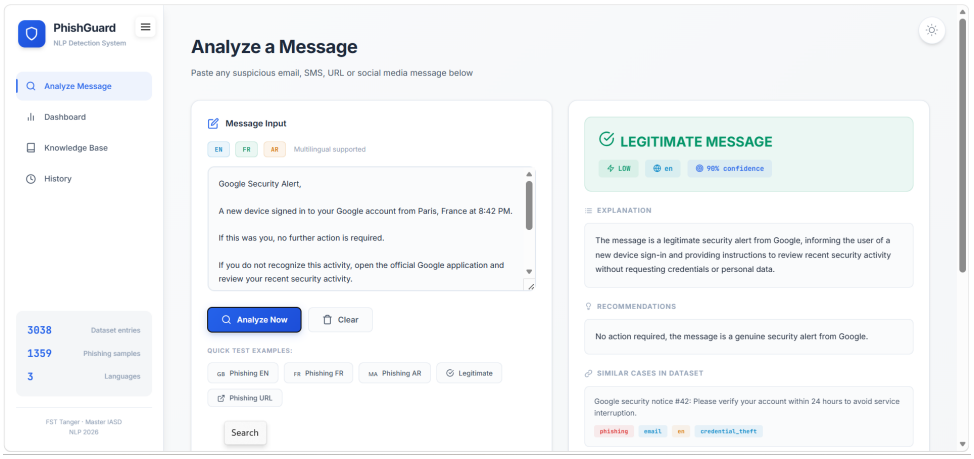

Figure 13: Message analysis result — legitimate message

8.2.2. Dataset Dashboard

Visualizes the dataset composition through:

- SVG donut chart for phishing/legitimate balance
- Animated bar charts for channel, language, and attack type distributions
- Four summary stat cards (total entries, phishing count, legitimate count, languages)

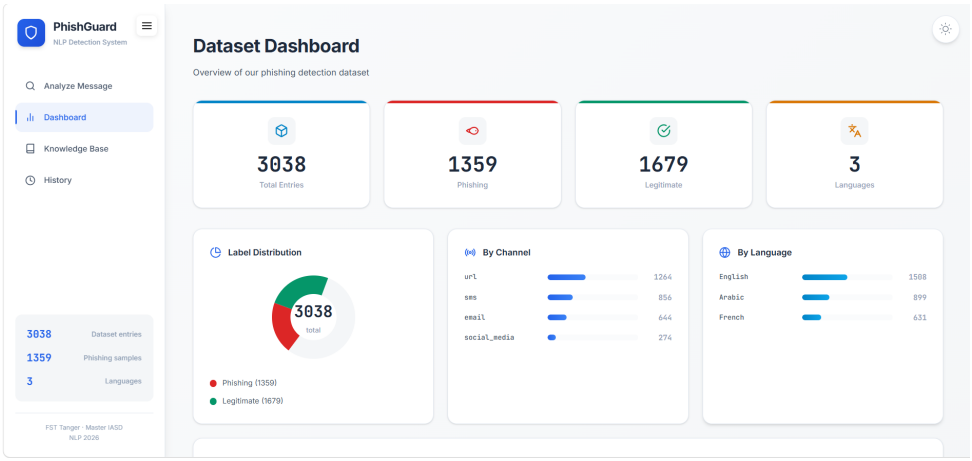

Figure 14: Dataset statistics dashboard

8.2.3. Knowledge Base

Allows dynamic extension of the RAG knowledge base:

- Drag-and-drop PDF upload
- Real-time indexing confirmation
- Persistent registry across sessions
- Active knowledge source display

8.2.4. Analysis History

Session-based history of all analyses with clickable reload functionality.

9. Results and Discussion

9.1. System Performance

The complete PhishGuard system achieves the following end-to-end performance characteristics:

Table 10: End-to-end system performance

Metric	Value
Retrieval Hit Rate@5	0.98
Retrieval MRR	0.899
LLM Accuracy (best)	77.0%
LLM F1-Macro (best)	0.766
Average search time	13 ms
Average LLM response time	2.15 s
Dataset size	3,038 entries
Languages supported	3 (EN, FR, AR)
Channels supported	4

9.2. RAG vs. Direct LLM

A key advantage of the RAG approach is grounding LLM decisions in concrete retrieved examples from the personal dataset. The retrieval context serves two critical roles:

1. **Evidence grounding:** The LLM cites patterns from real phishing examples rather than relying purely on parametric knowledge
2. **Explainability:** Retrieved similar cases provide transparent justification for end users

9.3. Limitations

- **French performance gap:** All models underperform on French (60% vs 82% EN for LLaMA-70B), attributable to dataset imbalance (631 FR vs 1,508 EN entries) and potentially weaker French phishing pattern coverage
- **URL liveness:** PhishTank URLs collected at time of data gathering may no longer be active, reducing the utility of URL feature extraction for archived entries
- **LLM dependency:** The system requires a live Groq API connection; offline operation is not currently supported
- **Benchmark sample size:** The 15-sample French benchmark subset is statistically insufficient for robust performance estimation

10. Conclusion

This work presents PhishGuard, a complete RAG-based multilingual phishing detection system built from the ground up according to the course constraints. All data was manually constructed and verified, no public datasets were used, and the LLM is exclusively integrated within the RAG architecture.

Key contributions include:

- A custom 3,038-entry multilingual dataset covering 4 channels and 3 languages
- A hybrid FAISS+BM25+RRF retrieval system achieving Hit Rate@5 of 0.98
- Benchmarking of 3 embedding models and 3 LLMs with quantitative metrics
- An integrated URL analysis module with 13 extracted features
- A complete web application with PDF knowledge base extension capability

Future work should focus on improving French language coverage, implementing offline LLM support, and extending the dataset with more diverse social media content.

References

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